Technical Report:
Link to Github Repository: https://github.com/rchivate/asteroids.git

Introduction

Space is filled with Near-Earth Objects (NEOs), but only a small fraction pose a legitimate threat to Earth. Distinguishing between a harmless flyby and a "Potentially Hazardous Asteroid" (PHA) is critical for planetary defense. Automated detection systems can act as an early warning system. By using NASA's real-world observational data, we can identify which physical characteristics (velocity, size, orbital proximity) most strongly correlate with hazardous designations.

Potentially Hazardous Asteroids (PHAs) are traditionally classified via deterministic orbital mechanics, defined by NASA's CNEOS as objects maintaining a Minimum Orbit Intersection Distance (MOID) ≤ 0.05 AU and an absolute magnitude (H) ≤ 22.0. However, a significant "characterization gap" persists; ESA estimates that approximately 70% of mid-sized asteroids remain unidentified due to sparse, noisy, or sunward-blind observational data. This research addresses the computational latency inherent in manual risk assessment by deploying a comparative machine learning framework (Logistic Regression, Random Forest, and XGBoost). By leveraging the NASA NeoWS API, we demonstrate that high-dimensional orbital feature spaces can be used to predict hazard status with near-perfect precision, even when physical characteristics are partially unobserved. This provides a scalable, automated alternative to traditional ephemeris-based screening, directly supporting time-critical planetary defense decision-making.

Through this project, the main question we are researching is if a machine learning model can accurately predict an asteroid's hazard status based purely on its physical and orbital characteristics. We explored some more specific and measurable questions to understand and evaluate this research question:

- To what extent do physical attributes such as the estimated diameter (minimum and maximum) predict hazard status?
- Which orbital features, such as velocity and miss distance, are the strongest indicators for determining risk?
- Can machine learning effectively identify the minority class of hazardous objects within a dataset dominated by safe ones?

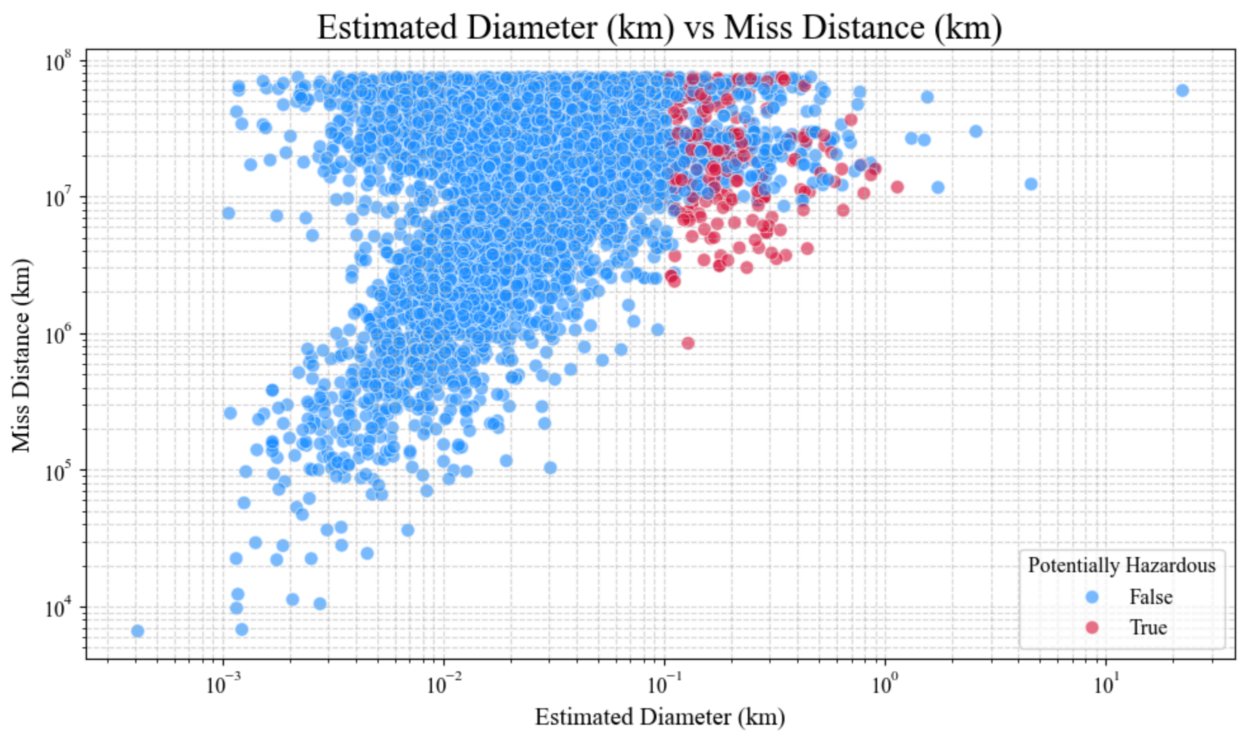
These insights are vital and can be very useful for space agencies and planetary defense initiatives by enabling them to prioritize tracking asteroids that pose the most threat to Earth.

Data

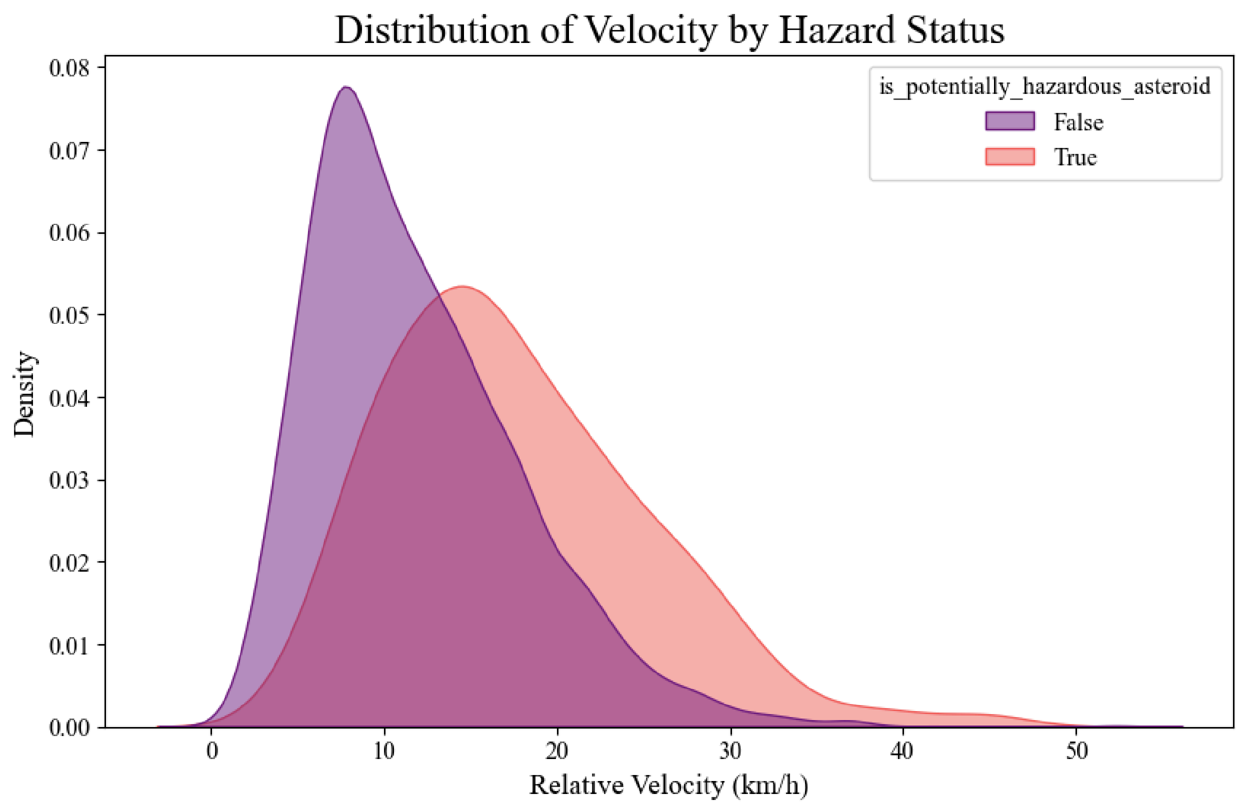
Preprocessing and Exploratory Data Analysis

During the preprocessing phase, we focused on data integrity and feature scaling. Any missing values or duplicates were removed to ensure data integrity and a clean input for the models.

- **Scaling and Encoding**: We performed scaling on physical dimensions and orbital distances to make sure features with larger ranges did not dominate the models.
- **Class Imbalance**: Initial analysis demonstrated a significant class imbalance since the majority of asteroids were not classified as hazardous. To address this issue, we used a train_test_split and prioritized Precision, Recall, and F1 Scores over accuracy, since it can be misleading when working with imbalanced datasets.



This graph showed an interesting relationship between Estimated Diameter, Miss Distance and an asteroid being classified as potentially hazardous. This graph clearly shows that these two variables, although they may be distinct indicators, are not the only indicators. There are other variables that factor into whether or not an asteroid is hazardous.



We looked into the relationship between estimated velocity and hazard status and found that there is a range of velocity where the frequency of hazardous asteroids is the highest.

Modeling

The data was split into 80% training and 20% testing sets. Three models were used to identify which model would be the best predictor to classify potentially hazardous asteroids:

1. **Logistic Regression**: Used as a baseline linear model to establish a minimum performance benchmark.
2. **Random Forest**: Used to reduce variance and capture non-linear relationships among orbital characteristics.
3. **XGBoost**: Used as a gradient boosting method to improve classification performance on complex datasets.

A Confusion Matrix was used to evaluate false positives and false negatives. Failing to identify a hazardous asteroid (false negative) is much more dangerous than identifying an asteroid as hazardous when it is not (a false positive).

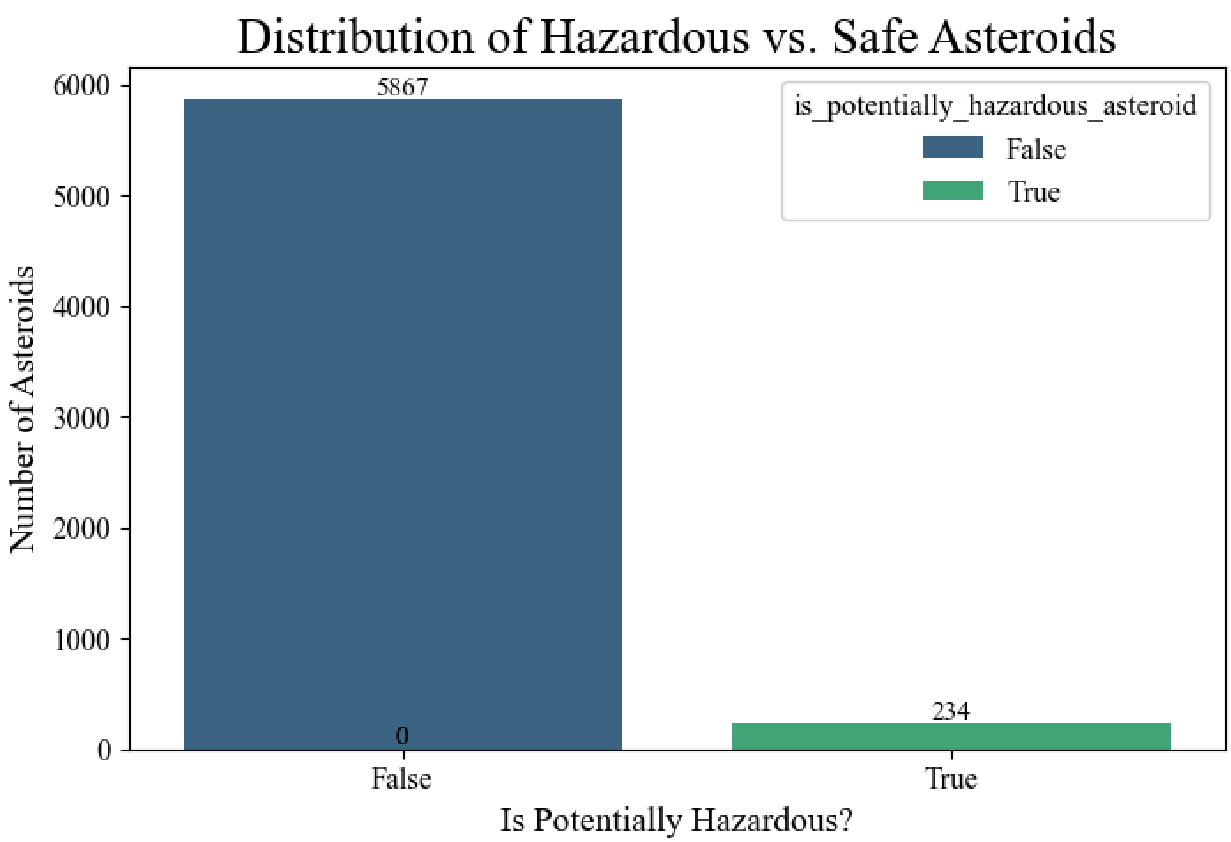
Results

The results demonstrated that XGBoost and Random Forest provide the highest Recall scores. Recall scores are a key metric measuring the model's ability to accurately and efficiently identify all hazardous asteroids. The Feature Importance chart indicates that the physical size (diameter) and miss distance of asteroids from Earth are the strongest predictors of classifying asteroids as hazardous. There is a strong correlation between the volume of an asteroid and how hazardous it is. The models demonstrate strong predictive power, however the limited features such as the missing data on asteroid composition (metal vs. rock), lower the accuracy of the dataset.

Conclusion

Limitations and Future Work

Some of the limitations that are involved with this dataset are that we have a limited number of variables and features provided. The lack of data on asteroid composition and density can make it difficult to interpret the impact risk of asteroids. "Unknown" variables in certain orbital tracks can impact predictive outcomes. Since the data only takes a look at data from the year 2025, the data might overfit to specific characteristics of the asteroids that occurred in that year. In addition, it could potentially miss several Potentially Hazardous Asteroids existing in different parts of their orbits. This could hide some patterns and skew the data.



The fact that there is not enough data about potentially hazardous asteroids since they are so rare is also a limitation because it makes it difficult for the models to learn about the characteristics of these types of asteroids.

Future Work on this project will involve implementing advanced XGBoost tuning to maintain high recall and improve precision. We will analyze historical impact data that involves a larger dataset including multiple years of data to identify long term risk patterns that can be used to improve planetary defense modeling.